Waiz Khan

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Research Interests: Foundation models; multilingual and multimodal representation learning; continual learning and memory; world models and robot learning; NeuroAI; scientific and medical machine learning.

Education

Johns Hopkins University Baltimore, MD
M.S.E. in Data Science *Aug. 2024 – May 2027*

- Selected coursework: Machine Translation; Artificial Intelligence; Computing for Applied Mathematics. Current: Methods in Computational Neuroscience; Natural Language Processing; Intro to Human Language Technology.

GD Goenka University Gurugram, India
B.Tech. in Computer Science & Engineering *Aug. 2019 – Nov. 2023*

Research Experience

Research Assistant Dec. 2025 – Present
Johns Hopkins University, Center for Language and Speech Processing | Advisor: Dr. Philipp Koehn *Baltimore, MD*

- Contributing to *Understanding and Improving Multilinguality of Large Language Models*, an NSF proposal led by Dr. Koehn and Kenton Murray; developed and stress-tested an intrinsic layerwise metric of multilingual representation structure across 17 open foundation models and up to 128 languages.
- Built the Slurm-based experimental and validation pipeline with fixed-effects/confound-adjusted analysis, preregistered confirmation and falsification tests, downstream validation, and frozen held-out-family evaluation; four of four blind predictions on unseen model families were confirmed.
- Ran controlled training interventions showing the representation structure is directly optimizable while naive optimization can trade off against transfer and capability, motivating constrained objectives; manuscript preparation is underway with Dr. Koehn.

Ancient Script Decipherment Research | ***Python, PyTorch, multimodal and graph learning*** 2026 – Present

- Rebuilt the project around leakage-controlled evaluation with provenance-aware ingestion across 12 corpora or inventories, 29 verified frozen partitions, grouped/near-deduplicated splits, test-access guards, configuration hashing, and an append-only experiment ledger; an Earlier Egyptian audit removed roughly 34% exact or near duplicates from candidate validation/test material.
- Built from-scratch visual and distributional baselines over a 640-type Egyptian sign inventory; a pilot label-free clustering study recovered partial known sign-function structure, motivating a stricter label-blinded held-out replication with bootstrap and permutation uncertainty.
- Current capstone work tests whether distributional, positional, visual, and graph-relational evidence improves blinded sign-function recovery; the graph arm compares Weisfeiler–Lehman/GIN structural features against degree-preserving rewired and shuffled null graphs, with cross-script alignment reserved as stretch work.

Research Assistant Aug. 2022 – Nov. 2022
GD Goenka University | Advisor: Dr. Yogesh Kumar; independently extended in 2026 *Gurugram, India*

- Studied chronic school absenteeism using IHDS-II national survey microdata and rebuilt the analysis over the full 47,027-student cohort; gradient boosting reached ROC-AUC 0.841 versus a 0.699 logistic-regression baseline.
- Used DoWhy to estimate the causal effect of school distance (ATE 0.14 per SD with refutation checks) and formulated a mixed-integer resource-allocation model that doubled expected impact under a fixed budget versus random targeting.

Experience

Founding Engineer & Co-Founder Apr. 2026 – Present
Arzaic LLC, Iris and Eve | Johns Hopkins Pava Center accelerator *Baltimore, MD*

- Architected Iris, an LLM agent platform for heart-failure care, as a staged gate → route → retrieve → generate → guard pipeline with semantic routing and agentic retrieval over a multimodal, versioned patient record.
- Built auditable record patches, multimodal clinical-artifact retrieval, a fail-closed safety judge, and an evaluation merge gate so system changes are safety-tested before deployment; automated care-plan revision drafts are produced for clinician review.
- Developing Eve as a medical-grounding research direction beginning with a verifier distilled from adjudicated safety labels, with longer-term self-supervised voice representations and longitudinal patient-state modeling.

Technical Skills

Programming & Systems: Python, C++, TypeScript/JavaScript, SQL, Bash; Slurm HPC, Linux, Git, Docker, ROS 2
ML & Research: PyTorch, Hugging Face Transformers, scikit-learn, CNNs/ConvNeXt, multilingual/multimodal and graph representation learning, RAG and agents; preregistration, held-out/walk-forward evaluation, bootstrap/permutation testing, DoWhy